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This is [Derek Abbott](#)'s guide to technical writing, primarily for use by his PhD students. But anyone is welcome to use this resource. Each PhD student is requested to go through this page as a checklist when writing a journal article or a thesis chapter.

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Introduction

Guidelines for technical writing are on this page. There are no rules for managing this page. Anyone can edit this page and add questions and discussion. It is self-moderating and you can delete and edit anything you like. To resolve

conflicts use the discussion page. If you come across your own hints and tips that you think are useful to others, feel free to add them.

Choice of writing package

- We use LaTeX because engineering is a highly mathematical topic, and this is the most professional way to present a mathematical paper. Also managing a very large document, such as a thesis, is actually easier in LaTeX. The overall quality of presentation beats Microsoft Word without question.
- Because LaTeX is a typesetting language, and not a word processor, it is the most efficient way for a journal to convert your paper into a final published form without making any mistakes in translating your mathematical expressions. If you send a highly mathematical paper in Word to a journal, the chances are that your maths will end up with many formatting errors that you then have to fix at the galley-proof stage.
- We use Microsoft Word for administrative matters, writing letters to editors, grant applications, CVs etc. But for technical journal articles and a thesis, LaTeX is the preferred medium. In some cases where we are collaborating on a journal article with non-engineers (eg. biologists) we will compromise and use Microsoft Word for ease of editing between many authors of different disciplines.
- For the thesis LaTeX style file download it from [here](#). You are strongly recommended to use this style file. Six postgrad alumni medal winning theses have used this style file and so it will give you a winning presentation style.

Spelling convention

- We use either American, British, or Australian spelling depending on the requirement of the publisher. In the case of a thesis, because this is Australia, we use Australian spelling. Also because we are electrical & electronic engineers we stick to the spellings "analog" (instead of "analogue") and "program" (instead of "programme"), even if your spell checker might sometimes tell you not to, simply because of the widespread use of these spellings in our field.
- In the case where two alternative spellings are allowed, eg. biassing or biasing, always chose the shortest one.
- There are cases where language is evolving and it is uncertain which spelling is out of date and which is acceptable as a modern replacement. For example, back in the 1990s it was unclear whether to write "infra-red" or "infrared." "Infra-red" was the old way to write it, but many people were starting to write "infrared." In order to decide, put both versions one-by-one into Google and see which one wins in terms of greatest number of hits. In this example, the spelling "infrared" has the most Google hits and is now clearly the dominant spelling.

Punctuation

- In Derek's group we follow all the standard rules of punctuation, except that for commas we obey the Oxford rule. The Oxford rule resolves ambiguity and it inserts a comma before "and" in the following situation:

Normal comma rule: Apples, oranges and bananas.

Oxford comma rule: Apples, oranges, and bananas.

- In a sentence where there are two commas, you can easily check yourself if the commas are in the right place. The trick is to read the sentence and skip the words between the commas. If the sentence still makes sense then the commas are correct.
- When inserting punctuation treat each mathematical equation as if it is a word. For example if an equation ends a sentence, the full stop must be on the end of the equation! Remember, equations obey the rules of punctuation as if they are like words. One equation, no matter how long, is like one word.
- Always put punctuation to the left of a closing inverted comma, not the right.
- Full stops, commas, colons, semi-colons, question marks, exclamation marks, all have no space on the left, but always have one space on the right.
- The dot at the bottom of a question mark or at the bottom of an exclamation mark serves the function of a full stop. So these must be at the end of sentences and there is no need to add another full stop.

e.g., eg. or eg

Where to put the dots? Well, it's one of those things that just depends on the style manual of the publication you are writing for. There's no real right or wrong here. In the UK, when I worked for the General Electric Company, in the 1970s we received a directive from the management that there is to be only one dot (seems they didn't have anything better to do). So since then, I have always written "eg." with one dot on the end in my own personal writing, unless dictated otherwise by the style manual of a publisher. I kind of like it because it means that when *et cetera* is abbreviated to "etc." it is then consistent with *exempli gratia* that is abbreviated to "eg."

Tone

The tone of your writing is important in order for it to be acceptable to a technical journal. Aim for your writing to:

- be formal
- be dispassionate
- avoid the passive voice, except when needed

Formal versus informal word usage

In scientific writing for journals and your thesis, you need to avoid informal words and use a formal equivalent. If you are writing a popular magazine article, then these requirements can be ignored. Here is a list of words to watch out for:

- Remove all contractions, as they are informal. For example, don't → do not, doesn't → does not, can't → cannot
- Do not use the first person, "I" in technical writing, or "my." "We" and "our" is okay.
- Avoid the possessive form for technical words: "the voltage's magnitude" → "the magnitude of the voltage"
- Amazing → significant
- Beautiful → elegant
- Big → large
- Bigger → larger
- Cheap → low cost
- Do → carry out; perform; conduct
- Done → carried out; performed; conducted
- Enormous → very large

- Fantastic → significant
- Get → obtain
- Great → significant
- Happen → occur
- Huge → very large
- Humungous → extremely large
- Marvelous → significant
- Some → a number of
- Splendid → significant
- Tiny → small
- Try → attempt
- Wonderful → significant

Dispassionate tone

Of course you must be completely passionate to be successful at your PhD. You need to be passionate about your research and passionate to share your exciting results by publishing. When we do technical writing the *tone* must be dispassionate, but you as a person can still be passionate! So what do we really mean by a dispassionate tone? We simply mean that you must remove emotional words from a scientific description. The science is about logic not emotion. So the emotional words have no place in your thesis or papers. You can use a little emotion in an acknowledgement section, because that is not the scientific part of the paper. When replying to reviews of your paper always remove any emotive words, even if you feel like strangling the reviewer.

Another exception is grant applications. A grant application has two parts: (a) a technical project description that is objective, and (b) a part that wants discussion about national benefit and significance, which is partly subjective. You must not use emotional words in the technical part. But in the subjective part you can get away with a few words that convey excitement, but don't overdo it.

Here's two examples of emotion in a technical paper and how to write them correctly:

- **Positive emotion:** "This is an exciting development." **Dispassionate version:** "This is a significant development because..."
- **Negative emotion:** "The measurement procedure is frustrating." **Dispassionate version:** "The measurement procedure has difficulties due to..."

Passive voice

If you overuse the passive voice your writing can sound cumbersome. When you minimize it the writing sounds more dynamic. To see a number of examples to understand the passive voice, see [here](#).

- Example when to avoid the passive voice

Passive voice: "The current is amplified by a transistor."

Active voice: "A transistor amplifies the current."

As you can see, the second option is much smoother and shorter.

- Example when the passive voice is useful

Passive voice: "The current is increased through the motor."

Active voice: "I increase the current through the motor."

Here the passive voice is useful because in a technical paper it is unimportant who actually increased the current. The passive voice keeps it general.

Grammar

Tense

In Derek's group we try to stick to the convention of keeping everything in the present tense, except when talking about previous work. For previous work you must use the past tense. This helps to keep the tone exciting and dynamic; long passages of past tense can sound boring. It also makes it easier for the reader to differentiate between the present work and any previous work.

An exception is when we are collaborating and publishing in a medical or biological journal. Here, these journals strictly want you to use the past tense for your experiment, but the present tense for your discussion and analysis. Physical and engineering journals accept present tense for your experiment, discussion, and analysis.

Split infinitives

A *split infinitive* is when you insert a word between "to" and its verb. For example:

Normal phrase: The spectrum analyser was used to study the device closely.

Split infinitive: The spectrum analyser was used to closely study the device.

Here, "study" is the verb and it has been split from the "to" in the second example. Split infinitives used to be thought of as bad grammar, but now in modern times many people accept them. However, some PhD examiners may be annoyed by them. Therefore the general advice is to minimize their use in formal and technical writing, where possible. However, feel free to use them occasionally when it makes the sentence clearer or flow better.

Words that are commonly misused

Here are some words in technical papers that commonly confuse people. Watch out for these cases.

Which or that?

When you are speaking "which" and "that" are interchangeable. This is not true for writing. The rules for using "which" and "that" in writing are as follows:

- Never insert a comma before the word "that." If you need a comma then remove "that" and replace it with "which."

- When the part of a sentence after "which" is explaining something about the sentence before the "which" then there is a comma before the "which." All other uses of "which" don't have a comma before them.

Resonance frequency or resonant frequency?

'Frequency' is a noun, 'resonance' is a noun, and 'resonant' is an adjective. An adjective describes a noun. If we say "resonant frequency" then we are describing the frequency with an adjective. But "resonant" is the description of the circuit, not the frequency. So the correct expression is "resonance frequency" which is a compound noun. So the whole expression is the name of the type of frequency.

Another example, to show what we have said makes sense is to consider the two nouns "fire" and "fly" (as in an insect). Putting the two together is then the name of a type of fly called the "fire fly." Notice we are not trying to describe the fly as being on fire, so we are not using any adjectives here. We are using two nouns and this is a compound noun that names the type of fly. This explains why "resonant frequency" is wrong, and "resonance frequency" is right.

Incidence angle or incident angle?

By analogy with the above argument for "resonance frequency," the correct term is "incidence angle." It is the electromagnetic radiation that is "incident" upon a surface not the angle itself. So the compound noun "incidence angle" is the name of the angle.

Antennas or antennae?

In electrical engineering, the plural of 'antenna' is 'antennas,' and not 'antennae.' Antennae are the little things on the heads of insects.

Neural or neuronal?

Strictly speaking, 'neural' relates to the nervous system, whereas 'neuronal' refers specifically to neurons. The nervous system does indeed contain neurons, but when specifically talking about neurons and you need an adjective always try to use 'neuronal' wherever possible. However, when the force of tradition has made a phrase mainstream, such as 'neural networks,' when referring to artificial networks of neurons, it is better to stick to what is commonly used.

Effect or affect?

Think of 'effect' (noun) as a phenomenon and 'affect' (verb) as an influence. Then you won't get confused. A trick for determining if you have used the correct word, is to replace it with the word with 'influence' in your mind. If the sentence still makes sense then 'affect' is needed. If the sentence sounds wrong then insert 'effect.'

Example when 'affect' is correct: The affect of the voltage... → The influence of the voltage... (sounds correct)

Example when 'effect' is correct: The Kerr effect... → The Kerr influence... (sounds wrong)

Envision or envisage?

They both mean the same thing, and mean to visualize or imagine. 'Envisage' is more common in British writing and 'envision' is more common in American writing. Use whichever suits your audience.

Various academic words

There are a number of academic words that you need to be aware how to correctly use:

- **Prove.** Avoid using the words "proof" and "prove" unless you are writing closed-form mathematics. In the physical world you cannot ultimately prove anything; we only increase confidence in models for given circumstances. Models are always subject to revision, especially when new measuring instruments open up a greater range of new circumstances. Therefore, for any physical discussion and experiments, always use "show" or "demonstrate" instead of "prove." On the other hand the mathematical world is not real, it is artificial. In maths you can make proofs as the mathematical space is well defined. In the physical world we do not know the extent of physical parameter spaces, as our instruments are not perfect. So we can never make closed form proofs in the sense that we can with mathematics. The physical world is always open-ended in that sense.
- **Law.** This is used in the sense of "Newton's Laws" for example. There actually is no such thing as a law in science. Use of this word is a mistake of tradition. All 'laws' named by humans are subject to revision. Therefore avoid using this term where you can. Only use it if it is already commonly used, such as in "Ohm's Law."
- **Ansatz.** This is a mathematical or physical starting guess, that you later verify by your results.
- **Paradox.** This is something that is true but appears to contain a contradiction. Most paradoxes can be resolved and it can be shown that the contradiction was not real in the first place. However, there are a few paradoxes that seem to have no solution. If you are talking about a paradox that has been solved, then to make this clear it is better to call it an *apparent paradox*. An exception is when referring to an apparent paradox by name; for example, "Simpson's paradox" is the name of an apparent paradox. It is just its name, and no one is suggesting that it is a real paradox. To say "Simpson's apparent paradox" is too long, and so we shorten the name for brevity.

Derek's golden rules

- Derek's **golden rule #1:** Never use the words "could," "would," or "should" in technical writing. "Could" and "would" are very weak sounding words, and "should" is too emotionally commanding. Suggested alternatives are:

Should → ought to, is possibly
Would → will potentially, will possibly
Could → may, can potentially, may possibly
- Derek's **golden rule #2:** Never begin a sentence with an abbreviation, mathematical symbol, or acronym. All sentences must begin with a real English word.
- Derek's **golden rule #3:** Nested brackets are forbidden. Although we allow nested brackets in mathematics, they are not allowed in the English prose.
- Derek's **golden rule #4:** Try to avoid using brackets wherever possible. Only use them when you have no other choice.
- Derek's **golden rule #5:** Try to avoid multiple-letter mathematical variables. Rather than inventing a multiple-letter variable, use a single letter with a descriptive subscript. This is always preferred. In cases were

multiple-letter variables have become tradition in the literature, such as SNR, then it is okay to use use them. But don't be guilty of making up new ones, because they are evil.

- Derek's **golden rule #6**: When referring to sections, chapters, equations etc., always be specific and refer to them by their number even if it is the next one. Don't be vague and say "In a later chapter we discuss..." but say "In Chapter 5 we discuss..."
- Derek's **golden rule #7**: Do not write about past events with relative time. Talk in terms of absolute time. For example, do not say "Last year, Smith *et al.* introduced a new extension to their algorithm." Instead of saying "Last year" write the actual year. Talking in relative time is not good practice in a thesis or a journal article. If someone reads your work in 20 years time, you still want your writing to make sense then.

LaTeX tips

- When a full stop ends a sentence, LaTeX automatically inserts a larger than normal space after it. So when you have a full stop with a space that is not ending a sentence, eg. in an abbreviation, always put a tilde (~) after that full stop. The tilde will then suppress the large space and it also suppresses a line break at that point. This is important, because you don't want a full stop at the end of a line if it is not the end of sentence.
- Don't get the three kinds of dashes mixed up in LaTeX: (1) - short dash or hyphen, (2) -- medium dash, and (3) --- long dash. Use the short dash for hyphenating words. Use the medium one for a number range. Don't use the short dash for a number range, as that is reserved for a hyphen. Use the long dash for a pause in a sentence (the pause is longer than a semicolon but shorter than a full stop). Another rule is that there is no space either side of these dashes. But there is a space either side of a minus sign.
- Only single-letter non-bold variables are in the italic maths font. Operators, numbers, constants, labels etc. are all in normal font. In a LaTeX equation if you need to force something to appear in the normal font use `\rm ...`.
- Physical units are always in upright font, so can be written outside the `$...$` environment.
- There is always a space either side of mathematical operations such as $+$, $-$, $<$, $>$ etc. The maths environment takes care of this for you. So always write `$a>0$` and not `$a$>0`, or the spacing will come out wrong. There are some exceptions to this spacing rule, for example, with minus signs in exponents; however, LaTeX will take care of the correct spacing for you. This is another good reason to always use LaTeX for technical writing.
- If you are inserting Matlab code or computer pseudo-code into a paper or thesis, use the `\tt ...` font.

Mathematics

- Use bold for vectors
- Single-letter non-bold variables are in the italic maths font. Everything else in the mathematics is in the regular upright Roman font. The backslash in front of functions such as `\sin` and `\cos` will take care of this for you, when using LaTeX. But, in other cases, to force the upright Roman font you need to use `\rm...`.
- Avoid using an asterisk to mean multiplication or convolution, in the main text. An asterisk is only allowed to mean multiplication in computer code. To mean multiplication, most of the time just put two variables next to each other. Sometimes you can insert `\cdot` if you want to be explicit. If you are multiplying raw numbers use `\times`. For convolution use `\otimes`. If you really need to use an asterisk for convolution, do not use `*` but use `\ast`. The `\ast` version uses the operator font and is properly centered.
- Remember each equation is considered part of a sentence and obeys the rules of English punctuation. In terms of punctuation, one equation counts as one word, no matter how long it is.

Spacing

- Full stops, commas, colons, semi-colons, question marks, exclamation marks, all have no space on the left, but always have one space on the right.
- There is always a space between a number and the units. Normally you must create the space with a tilde, in LaTeX, to suppress a linebreak.
- Note the special case that a number has no space after it, if followed by a percentage sign or a degree symbol.

References

- When writing a journal article, always follow the format required by the specific journal.
- When writing your thesis always use Harvard style referencing. This means you cite using author name and year, rather than by a reference number. For a huge document such as a thesis, the Harvard style is preferred by examiners because they can often recognize a reference from the name and year. This saves them from constantly having to check the references at the end. Also Harvard style is much easier to manage in a large document, because the references are all at the end alphabetically. In LaTeX, use `\cite` as normal, and it is the style file that will create the Harvard style. However, when you refer to the reference itself as a noun you must use `\citeasnoun`.
- For the formatting and style of a reference, you follow the rules of the journal. If you are writing a thesis, because we are electrical & electronic engineers, format it the IEEE way. Simply refer to any published IEEE paper and follow the way they do it.

How to annoy the hell out of a PhD examiner

Prof. Bob Bogner used to have a list of ways to annoy a PhD thesis examiner. These also apply to journal reviewers. Remember, this is a list of what *not* to do. So make sure you avoid them. I have adapted his original list as follows:

1. **Use the word 'this' ambiguously.** "The dynamic system goes unstable when the feedback coefficient exceeds unity and then the machine may over-speed and explode. This may be avoided by the use of the proposed design method." Notice that "This" could refer to the system, to the instability, to the over-speed, to the explosion, or to the situation. Be aware that the word "it" also can have the same ambiguity problem.

2. **Use "however" as a conjunction.** "The coil overheats, however the motor does not catch fire." Try to avoid this. Three correct alternatives are:

- "The coil overheats, and the motor does not catch fire."
- "The coil overheats, but the motor does not catch fire."
- "The coil overheats. The motor, however, does not catch fire."

3. **Use "so" as a conjunction.** "The train left early so many people missed it." The correct connection is made by the use of "and so", indicating the cause and effect relationship: "The train left early and so many people missed it." A different sense is implied by the construction: "The voltage is low so that users will be absolutely safe." This case is allowed because "so" indicates a purpose for the voltage being made to be low.

3. **Use "it's".** The word "it's" with an apostrophe should not appear in technical writing. We do not allow contractions, and so its use to mean "it is" is not allowed. The case of "its" without the apostrophe is the possessive form. So you use it correctly as follows: "The resistor is connected, and its function is to introduce a voltage drop."

4. **Missing or erroneous references to figures.** Make sure every figure is referred to in the text.

5. **Omit the object of comparison.** "This thesis presents a better method for designing amplifiers." Better than what?

6. **Confuse singular and plural words.** Tricky cases to watch out for are:

- Abscissa vs. abscissae
- Antenna vs. antennas (not antennae for engineering!)
- Appendix vs. appendices
- Automaton vs. automata
- Axis vs. axes
- Criterion vs. criteria
- Corpus vs. corpora
- Datum vs. data
- Die vs. dice
- Emphasis vs. emphases
- Equilibrium vs. equilibria
- Extremum vs. extrema
- Focus vs. foci
- Formula vs. formulae or formulas
- Hypothesis vs. hypotheses
- Index vs. indices
- Locus vs. loci
- Matrix vs. matrices
- Maximum vs. maxima
- Medium vs. media
- Minimum vs. minima
- Modulus vs. moduli
- Nucleus vs. nuclei
- Optimum vs. optima
- Phenomenon vs. phenomena
- Polyhedron vs. polyhedra
- Quantum vs. quanta
- Radius vs. radii
- Radix vs. radices
- Schema vs. schemata
- Simplex vs. simplices
- Spectrum vs. spectra
- Stratum vs. strata
- Synopsis vs. synopses
- Synthesis vs. syntheses
- Thesis vs. theses
- Vortex vs. vortices

7. **Treat singular concepts as if they have gradations.** "The parameters have an improved level of optimality." Words such as "unique", "optimum", etc. do not have gradations. Something is unique or it is not. Something is optimum or it is not. It is like virginity.

8. **Use the word "hope" or "hopeful".** These types of words are meaningless in scientific writing and are completely banned. Do not use them at all. "We hope our model is an improvement over Shannon's model" should be "Our model is aimed at providing an improvement over Shannon's model." Also, "Hopefully the voltage will not be excessive" implies that the voltage hopes like a person. A better form is: "It is anticipated that the voltage will not be excessive."

9. **Use capitals incorrectly.** This can really annoy the reader. Never use block capitals for emphasis, as it looks very uneducated. Don't use capitals for chemical compounds, unless at the beginning of a sentence. The rules for capitals in English text are:

- Capital at the beginning of a sentence
- Capital for proper nouns (i.e. named things)
- Capitals for acronyms
- Capitals for Section 1, Figure 1, Equation 1, Table 1 etc, because you are naming those items. When you are generally referring to sections, figures, equations, tables etc. they are lower case because you are not naming a specific one
- Capitals in title case for any part of a reference that is in italics
- Capitalize the first letters of all German nouns when referencing a paper in the original German

10. **Omit the subject of a participle.** "Making the insulation thin the strength is too low and it may rupture." Who or what is making the insulation thin? The sentence construction implies that the strength made it thin. Perhaps the author meant "Making the insulation thin causes the strength to be too low and it may rupture." Who can tell what the author meant if the words are ambiguous?

13 **Use "/" in written text to mean "or".** This is sloppy and is banned.

14. **Use the phrase "as such".** This leads to sentences that sound like they contain superfluous waffle and is best avoided.

Why all the above rules are important for your PhD thesis

On a trip to the University of Cambridge, Ken Moxham from our Civil Engineering department once saw the following passage: "Because of the impracticability of an examiner's establishing the details of the conduct of the research, the care with which a thesis is prepared will be taken as *prima facie* evidence of the care with which the research has been carried out."